

the imaginary brake anytime you are sitting in the car with someone else? No matter how much you trust the driver, or how familiar you are with his/her driving, you cannot help but feel that sense of panic whenever you feel them racing to make it through the green light. Now imagine the same situation, only this time there is no driver and you are on the passenger side. No matter how excited you might be about self-driving cars, it is going to be a hard adjustment to make.

2. We are not educated about autonomous driving. There is a reason why most of the tech giants want to offer self-driving cars as a cab hailing service. Because that way, you don't have to spend any time learning how to handle the car. You simply order a cab using an app. You get in and get out upon reaching your destination, and that's it. However, owning a self-driving car is a whole different scenario. There are a whole bunch of safety features that you will have to learn to use. You will also have to get educated and trained to handle emergencies that could occur with the car. It is a technology, so it will fail you from time to time. And, in the early days, the snags are going to be even more. It is not going to be a picnic to own a self-driving car.

3. We don't have the money. Self-driving cars cost ... A LOT. The sensors, lidar, cameras, and a whole bunch of other sensors and technology that will have to go into it will massively increase the cost of owning a self-driving car. This further puts a hurdle between these cars becoming an everyday phenomenon on the streets. For self-driving cars to become common, they will have to be built keeping in mind what a common man can afford, not just in terms of purchasing a vehicle but also maintaining it. Self-driving cars cannot just be for the elite, like owning a Tesla.

4. We don't trust technology. If you are following the news right now, then you are aware of the fiasco that is the Covid vaccine rollout in any country that has the vaccines. A huge part of it is due to mismanagement by the governments. But a significant problem is also the lack of trust that people have in medical technology when it comes to their own health.

What is the correlation in the vaccine rollout and self-driving vehicles? A major one. People don't trust technology, especially when it is made by the industry giants. If a substantial part of the population is not willing to get a vaccine to save their lives, will they be willing to put their lives into the hands of a self-driving car built by big tech? Not everyone is thrilled about the self-driving car. And these concerns are genuine. A huge media campaign is required to help people build public trust in this technology. And this can take a long time.

5. We don't have the infrastructure. Most countries in the world do not have the right infrastructure to support self-driving cars. The technology is good, but it is still far from being able to drive on unmapped dirt roads in the mountains or deserts. Additionally, network connectivity is an ongoing issue in the entire world. Even developed countries such as the USA have many issues with network connectivity. For self-driving cars to be fully functional, we require solid, uninterrupted 5G network connectivity throughout the country and wherever else we may choose to go by car. We are still way behind achieving this goal.

6. We don't have the rules. A complete overhaul of transportation legislation is required in every country. And this is not a one and done thing. Overhauling of the legislation will be an ongoing process for at least a good twenty years after self-driving cars arrive. Why? Because we are going to learn from our mistakes. The legislation designed the first time will inevitably have many issues. It is going to have to be altered many times before we get it right. There is also the fact that self-driving cars cannot arrive in the scene before we do this. A complete change of driving rules is a necessary, albeit tough, prerequisite to autonomous vehicle technology.

What is the future?

I am not a 'tech swami.' I cannot give you a definitive answer as to when the self-driving car will 'arrive in the scene.' But I can give an answer that is rooted in probability and research. I understand that there is a shared sense of lethargy and pessimism in the

public opinion about self-driving cars right now. What was a 'big deal' in 2010 doesn't really seem as such right now.

But a self-driving car cannot and should not be equated with a piece of technology like 'Google glass.' Self-driving cars are here to stay. But the transition isn't going to be as fast as was the case with smartphones. Instead, this is going to be a slow and painful transition in the case of most countries. There are going to be a lot of technological and social snags that will have to be dealt with before this technology completely takes over.

However, as far as the technology behind autonomous vehicles is concerned, the self-driving car is most definitely the next step in the evolution of the automobile industry. Experts currently put the estimate at the year 2050 in order to see a massive reduction of up to 90% in road accidents caused as a result of driving. Is the estimate too optimistic? I don't think so.

From the perspective of an AI researcher, I feel safe enough to say that the one thing that we don't lack is data. We have a lot and I mean A LOT of data to better our deep learning models. We are in the golden period of AI, and although AI is already being used everywhere, we are just seeing the beginning of its potential. With the speed with which AI research is moving, a lot of the challenges I mentioned previously can be taken care of by using AI. For instance, we don't need a complete overhaul of our infrastructure if the AI learns to handle driving in the current conditions. AI can also help in reducing the cost of the technology by being able to get the job done with lesser resources.

The sheer power and potential of AI makes me highly positive and optimistic about autonomous vehicle technology. And while we might go through many phases of transformation before we are able to call this technology 'mature,' we certainly can and will get there. **EFY**

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